Contents

Why Numerical Methods? 9

Floating-Point Arithmetic & Rounding Errors 11

Conditioning & Stability 13

Fixed-Point Iteration: Concept & Convergence 15

Speed of Convergence & Acceleration 17

Newton Method (1D) 19

Newton Method for Systems 21

Robust Root-Finding Methods 23

Why Numerical Integration? 25

Newton–Cotes Formulas 27

Gaussian Quadrature 29

Polynomial Interpolation 31

Splines & B-Splines 33

Fourier Series & Approximation 35

Method Selection & Numerical Thinking 37